

Main result. Translated bias scores can be meaningful behavior even when the hidden route is not representation-equivalent: non-English prompts may pass through an English hub that CKA misses.

What Translated Benchmarks Assume

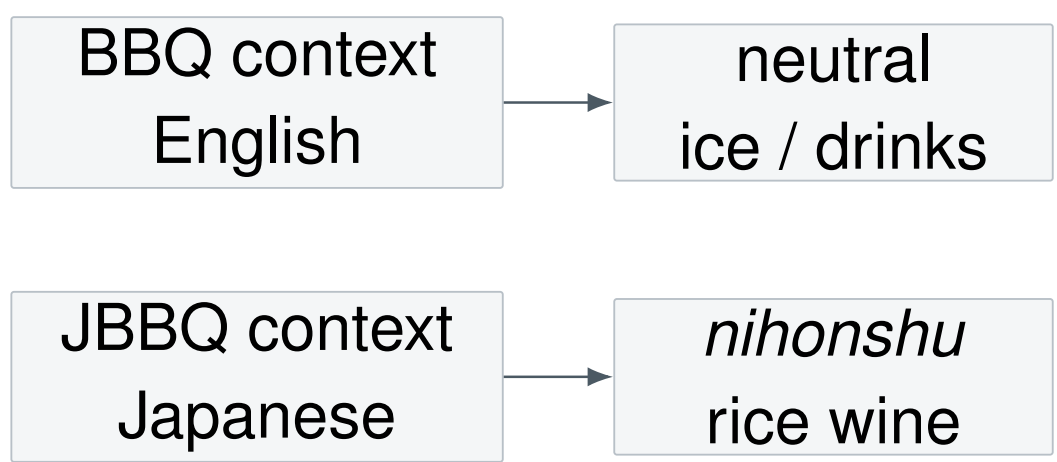
Translated bias benchmarks score stereotype preference after translating an English BBQ item into another language. This treats translation as construct preservation.

Question. Is a target-language bias gap behavior, or English-hub routing?

Model	Training regime
Llama-2	English-centric base
Swallow	Llama-2 + Japanese adaptation
koen	Llama-2 + Korean adaptation
LLM-jp	balanced Japanese/English pretraining

Matched pairs fix architecture and initialization; changes mostly reflect adaptation.

Same Event, Different Continuation



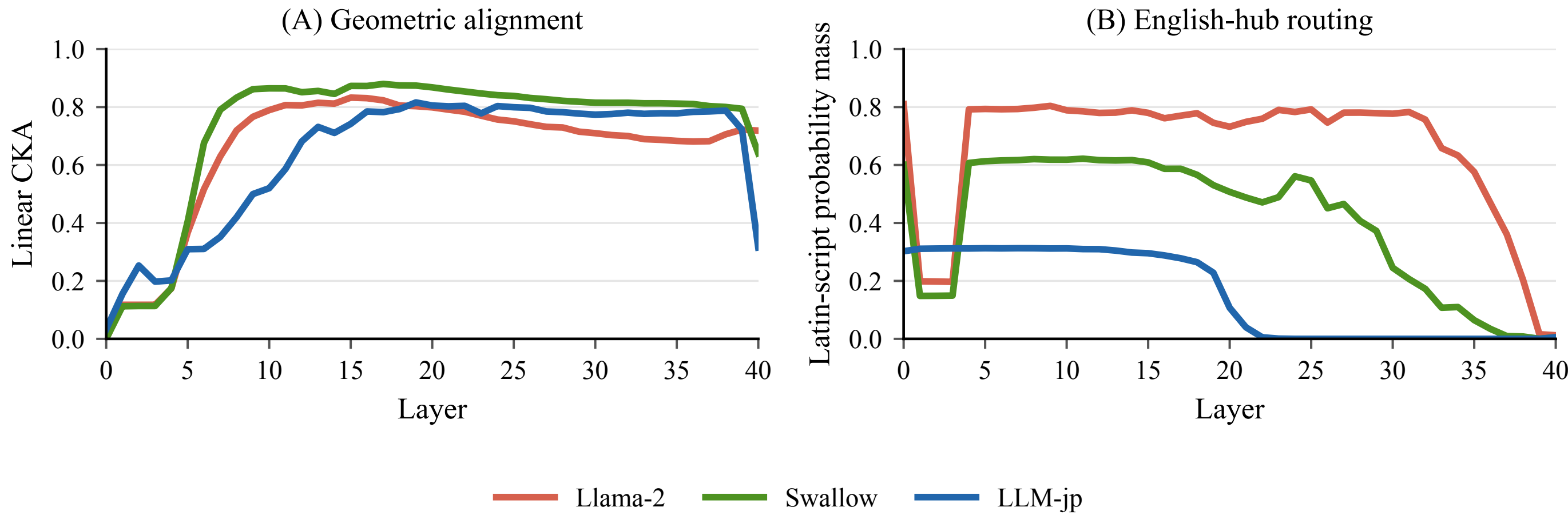
The event is preserved, but the continuation can become culture-specific.

Four Checks for Comparability

CKA	geometric alignment
Script-mass lens	predicted token script
Binned lens JSD	semantic divergence
Hub patching	causal transplant test

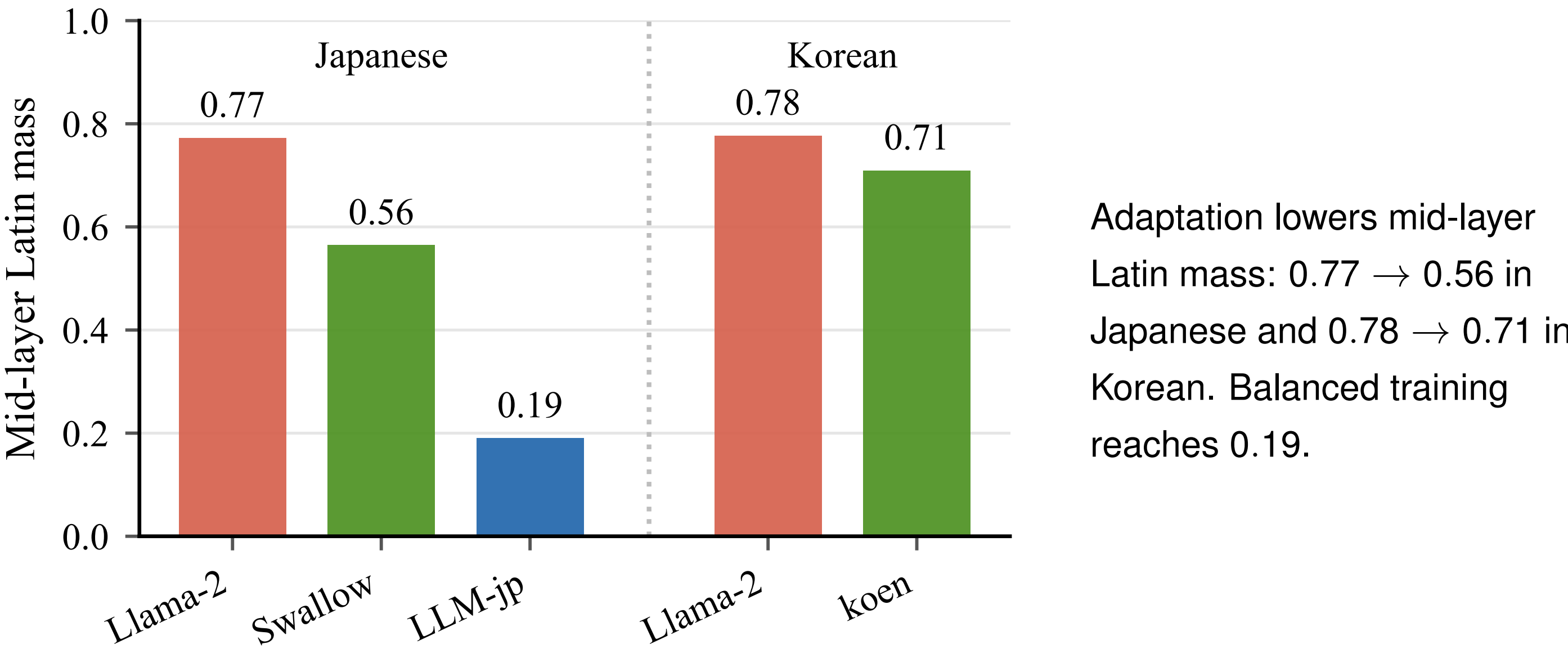
Representations use context-only hidden states; behavior uses full question-answer items.

CKA Hides English-Hub Routing

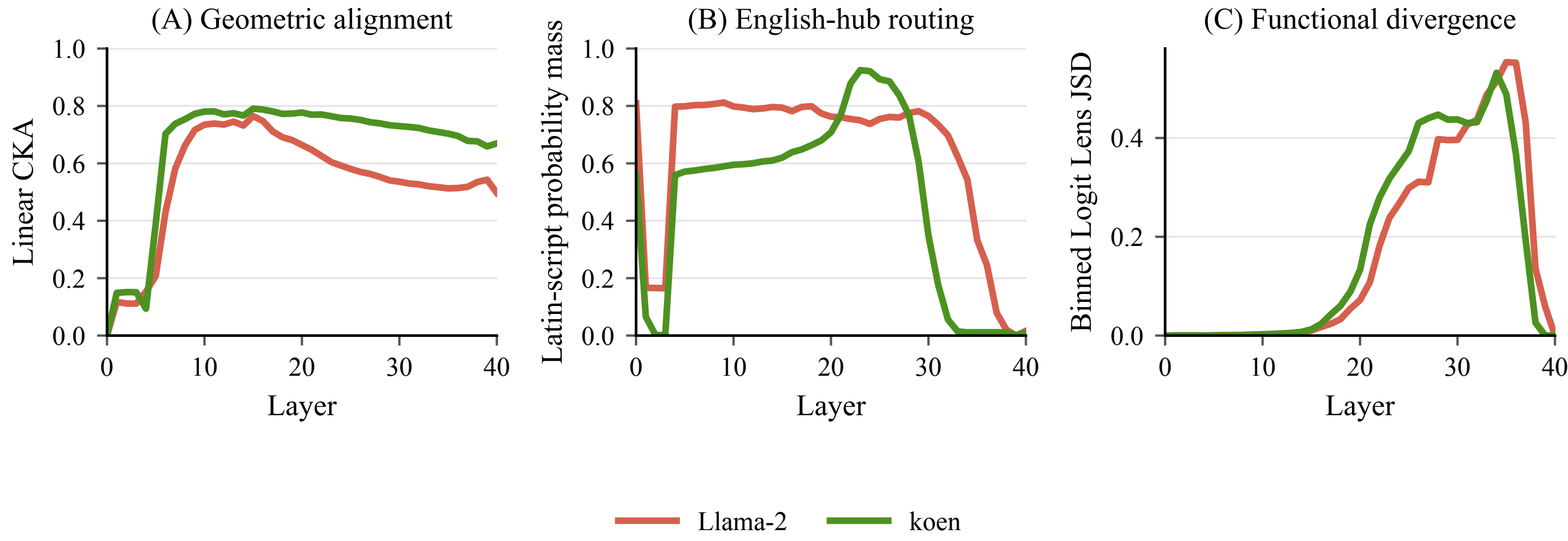


CKA suggests alignment, but the script-mass lens shows Llama-2 predicts English-script tokens in Japanese middle layers.

Target-Language Adaptation Reduces Hub Routing



English-Korean Routing Replication



English/Korean reproduces the diagnostic split: sustained CKA, reduced English-script mass, and late functional divergence.

Bias Behavior Is Language-Specific

Japanese behavior			
Model	EN	JA	Diff
Llama-2	0.487	0.344	+0.142
Swallow	0.496	0.352	+0.144
LLM-jp	0.473	0.343	+0.130
Korean behavior			
Model	EN	KO	Diff
Llama-2	0.562	0.508	+0.053
koen	0.507	0.541	-0.034

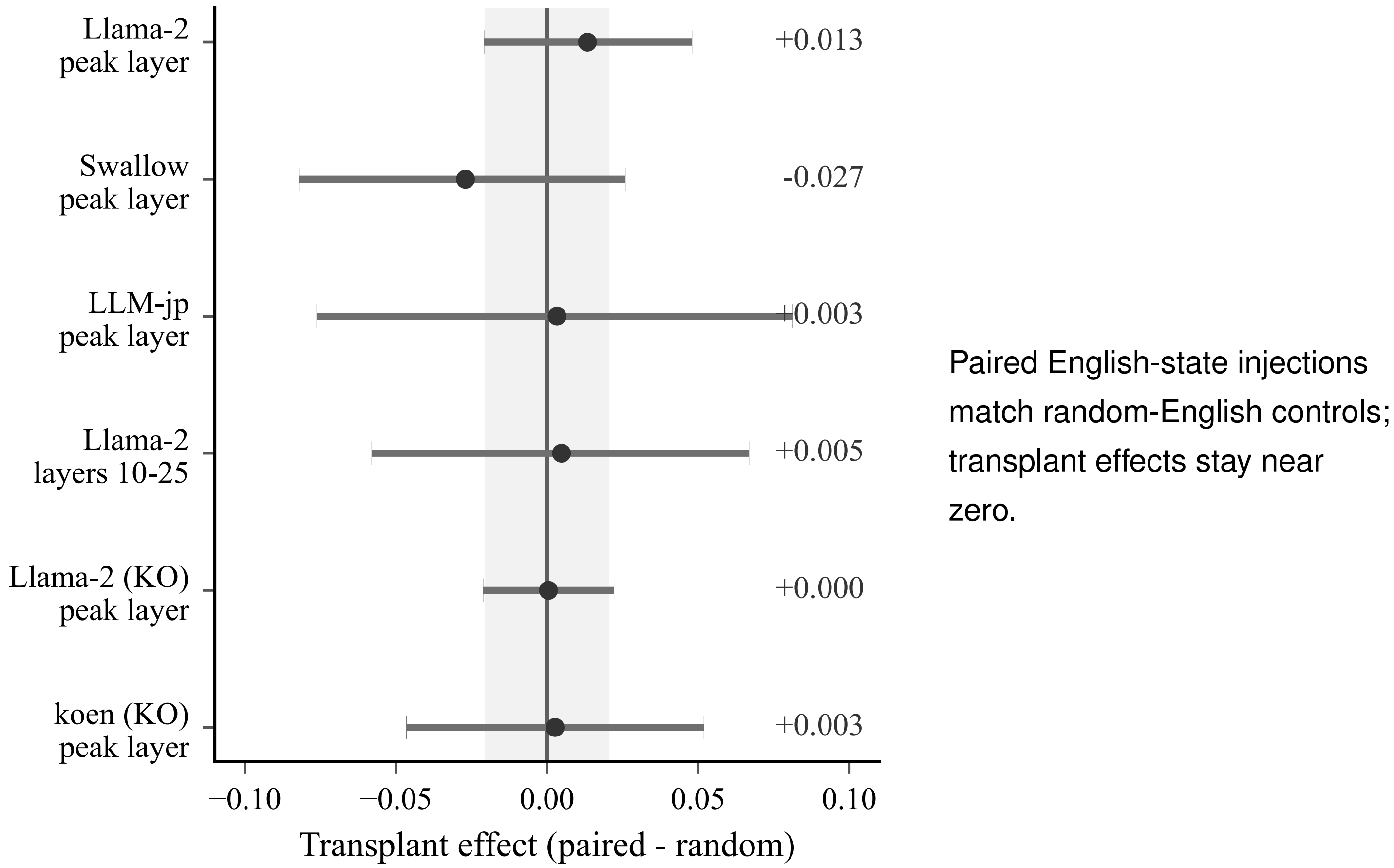
P_{ster} in ambiguous items. Japanese shows a robust English-over-target gap; Korean is closer to English.

Japanese Gap Varies by Category

Category	EN	JA	Diff
Age	0.491	0.305	+0.186*
Disability	0.477	0.305	+0.172*
Gender identity	0.577	0.446	+0.132*
Physical appearance	0.462	0.463	-0.001
Sexual orientation	0.444	0.271	+0.173*

* bootstrap 95% CI excludes zero. Physical appearance is the exception.

Hub Routing Does Not Transplant Bias



Dissociation and Audit Protocol

Representations: translated prompts are not necessarily processed in comparable spaces.

Behavior: per-language bias scores remain meaningful; patching finds no transplant of English bias.

1. Check tokenizer feasibility.
2. Pair CKA with a script-level lens.
3. Interpret scores as behavior.
4. Filter translations with surface-form chrF.

Limitations: Korean is a directional replication ($n = 261$); patching is a strengthened null. References: Parrish et al. 2022; Yanaka et al. 2025; Jin et al. 2024; Neplenbroek et al. 2024; Wendler et al. 2024; Kornblith et al. 2019.